

Alyson Souza Carregosa

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Education

Maurício de Nassau University

Associate Degree in Systems Analysis and Development

Jan 2023 – Dec 2025

GPA: 9.38/10 Aracaju, Sergipe

Estácio

Bachelor's Degree in Computer Science

Jan 2025 – Dec 2027

GPA: 9.39/10 Aracaju, Sergipe

Prof. Experience

Support Informática

Backend Developer

On-Site

Jan 2025 – Present

- **Performance Engineering & Platform Rewrite (Node.js → .NET):** Led the full backend migration of *Saef Comanda* (company's flagship product) from Node.js to .NET, reducing data traffic by **41%** (payload optimization from 8MB to 4.7MB) and improving request latency by **35%**.
- **Business Impact & Scalability:** The new C# architecture enabled approximately **43% growth** in daily sales volume processed by clients, supporting high-demand operations such as [Pandoro](#) and [Jardins Delicatessen](#).
- **Ecosystem Development & Integrations:** Contribute to the Saef ecosystem including [Administrative Dashboard](#) and [Sales Portal](#). Built integrations with payment gateways (Pagar.me), CRM (Bitrix24), and currently leading e-commerce connectors (Mercado Livre and Shopee).
- **Legacy Modernization (Delphi → C#):** Key contributor in migrating legacy systems to .NET, standardizing APIs and improving overall maintainability and performance.

Escariz

Backend Developer

On-Site

Aug 2024 – Dec 2024

- **Marketplace Integration Architecture:** Developed APIs for bidirectional synchronization of catalog, inventory, and orders with Amazon, Mercado Livre, and Shopee — transforming the internal platform into a centralized sales hub.
- **Data Engineering & Enrichment Pipelines:** Implemented metadata processing pipelines (Book Info) and integrated with market intelligence tools (Keepa) to automate product data enrichment and ranking optimization.
- **Database Performance Engineering:** Managed complex SQL Server environments, optimizing queries and routines to support high-traffic e-commerce workloads.
- **Legacy Optimization:** Led performance refactoring initiatives that reduced loading bottlenecks and improved end-user experience.

Github

[EventFlow](#) | C#, .NET, Redis, Docker, Serilog, OpenTelemetry, SQL Server, xUnit

March 2025

Designed as a real-world backend architecture simulation, EventFlow explores performance, observability, and system resilience under production-like conditions.

- Implemented Redis using the **Cache-Aside pattern** to address read bottlenecks and reduce database pressure, achieving millisecond-level response times.
- Introduced full **observability** (OpenTelemetry + Jaeger + structured logging) to expose hidden latency between API and database layers — focusing on operational transparency rather than just functionality.
- Containerized the full stack via **Docker** Compose to emulate distributed environments and validate infrastructure-level decisions.
- The project was intentionally built to study trade-offs between performance optimization and architectural complexity.

Focus: Performance Engineering, Observability, Distributed Design.

Finance API was created to explore separation of concerns and scalability patterns in financial domain modeling.

- Applied **CQRS** and **Mediator** to isolate write/read responsibilities, improving maintainability as business logic grows.
- Introduced **distributed caching** strategies to mitigate database hotspots in read-heavy operations.
- Implemented secure authentication using **JWT** and **ASP.NET Data Protection**, focusing on real-world deployment concerns such as key persistence in containers.
- Emphasis was placed on **clean boundaries between layers**, reflecting enterprise-grade architectural practices.

Focus: **Architecture Evolution, Security, Maintainability.**

CatalogHub explores scalability challenges when handling product media and relational data simultaneously.

- Designed storage abstraction separating relational persistence (**PostgreSQL**) from blob storage (**AWS S3**), simulating real-world cloud architectures.
- Implemented standardized API response patterns and robust pagination to improve client-side consumption and consistency.
- Leveraged LocalStack to emulate **AWS Services** locally, enabling realistic cloud workflows without external dependencies.
- The project emphasizes infrastructure thinking beyond CRUD APIs.

Focus: **Cloud Architecture, Data Modeling, API Consistency.**

Technical Skills

Languages & Frameworks: C#, .NET, ASP.NET Core.

Architecture & Design: Clean Architecture, SOLID, DDD Concepts, CQRS, Mediator Pattern, Microservices.

Observability & Performance: OpenTelemetry, Serilog, Distributed Tracing, Redis.

Messaging: RabbitMQ, Azure Service Bus (concepts).

Databases & ORM: SQL Server, PostgreSQL, Entity Framework Core, Dapper.

Security: JWT Authentication, Authorization Policies, OAuth 2.0.

Testing & Quality: xUnit, Moq, SonarQube, CI/CD.

Cloud & DevOps: Docker, AWS, GitHub Actions, Git Flow.

Languages: English (B2), Spanish (A2), Portuguese (Native).

Leadership & Community

Organizer

- **Leadership and Community:** Organize technical events and workshops fostering the local tech ecosystem.
- **Networking & Collab:** Coordinate speakers, partnerships, and community growth initiatives. Collaborated with companies such as [Jetsales](#), [Quero Delivery](#), [Fwk](#), [Avalue Ventures](#), [Get Info](#) and [Alura](#).